

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 12 Higher School Certificate
Trial Examination Term 3 2017

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- A reference sheet is provided separately
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room.

Total marks – 70

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet
provided

Section II Pages 7 – 13

60 marks

Attempt Questions 11 – 14

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	Total
Total	/10	/15	/15	/15	/15	/70

This assessment task constitutes 45% of the Higher School Certificate Course School Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

1 What is the domain of $y = \sin^{-1} \frac{x}{2}$?

(A) $-\pi \leq x \leq \pi$

(B) $-2 \leq x \leq 2$

(C) $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

(D) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

2 A particle is moving in one dimension. Given that $v^2 = 4x^2 - 8$, where x is displacement in metres and v is velocity in metres per second, the acceleration of the particle at 2 metres right of the origin is:

(A) 0 ms^{-2}

(B) 2 ms^{-2}

(C) 4 ms^{-2}

(D) 8 ms^{-2}

3 The number of followers of an Instagram account (N) grows according to the formula $N = 80 + 20e^{kt}$, where t is time in hours and $k = 0.2$.

How long will it take for the Instagram account to have 1000 more followers than it started with?

(A) 19 hours 8 minutes

(B) 19 hours 34 minutes

(C) 19 hours 40 minutes

(D) 20 hours 6 minutes

- 4 Which group of three numbers could be the roots of the polynomial equation $x^3 + 9x^2 + 8x + d = 0$?
- (A) $-6, -5, 2$
(B) $-20, 6, 5$
(C) $-3, -3, -3$
(D) $-1, -2, -6$
- 5 The acute angle between the lines $3x + 2y - 7 = 0$ and $2x - y + 1 = 0$ is closest to:
- (A) 14°
(B) 41°
(C) 60°
(D) 76°
- 6 What is the derivative of $2\cos^{-1} 3x$?
- (A) $\frac{2}{\sqrt{1-9x^2}}$
(B) $\frac{-2}{\sqrt{1-9x^2}}$
(C) $\frac{-6}{\sqrt{1-9x^2}}$
(D) $\frac{6}{\sqrt{1-9x^2}}$
- 7 Let $|b| \leq 1$. What is the general solution to $\cos \frac{x}{2} = b$?
- (A) $k\pi \pm \cos^{-1} b$
(B) $2k\pi \pm \cos^{-1} b$
(C) $4k\pi \pm \cos^{-1} b$
(D) $4k\pi \pm 2\cos^{-1} b$

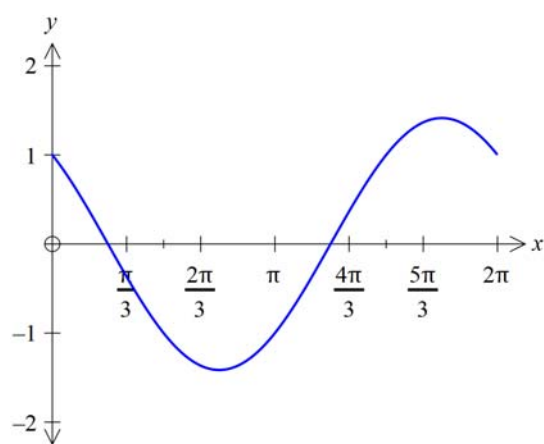
- 8 The point P divides the interval $A(-2, -2)$ to $B(6, 4)$ externally in the ratio $1:3$.

The y -value of P is:

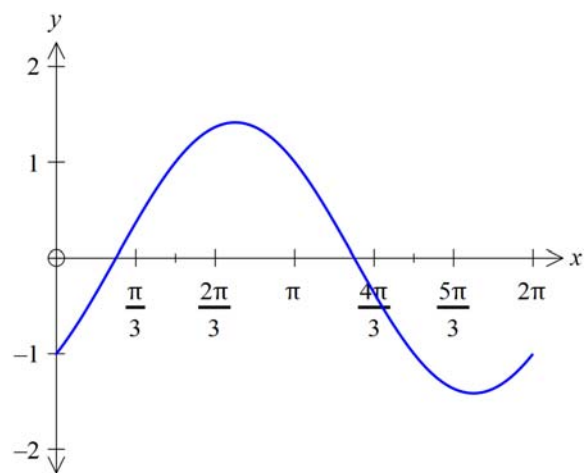
- (A) -5
- (B) -1
- (C) 1
- (D) 2

- 9 The graph of $y = \cos x - \sin x$ for $0 \leq x \leq 2\pi$ is:

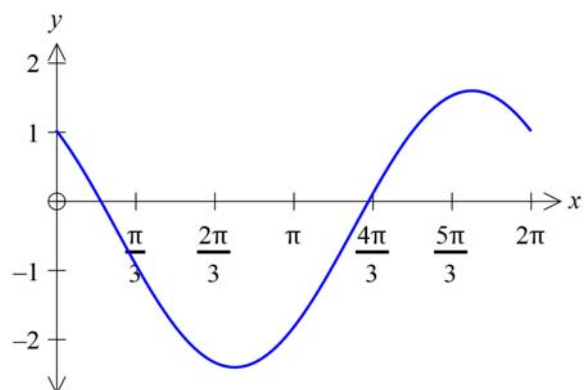
(A)



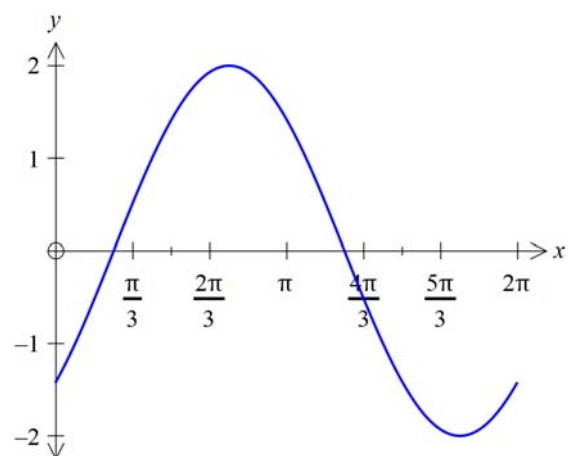
(B)



(C)



(D)



10 The primitive function of $\frac{1}{x^2 + 2x + 3}$ is:

(A) $\log_e(x^2 + 2x + 3) + C$

(B) $\frac{1}{2} \tan^{-1}\left(\frac{x+1}{2}\right) + C$

(C) $\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right) + C$

(D) $\frac{1}{4} \tan^{-1}\left(\frac{x+1}{4}\right) + C$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

(a) Solve $\frac{2}{x-1} \geq x$. 3

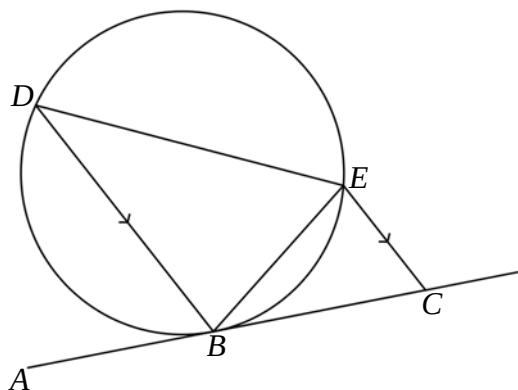
(b) Find the exact value of $\int_0^{\pi} \cos^2 3x \, dx$. 3

(c) (i) Expand $\left(x + \frac{2}{x}\right)^5$, giving your answer in simplest form. 2

(ii) Hence, simplify $\left(x + \frac{2}{x}\right)^5 + \left(x - \frac{2}{x}\right)^5$. 2

(d) Evaluate $\int_0^1 \frac{e^x}{\sqrt{e^x+1}} \, dx$, using the substitution $u = e^x$, giving your answer in exact form. 3

(e) In the diagram below, AC is a tangent to the circle at B and $BD \parallel CE$. 2



NOT TO
SCALE

Prove that $\angle BCE = \angle BED$

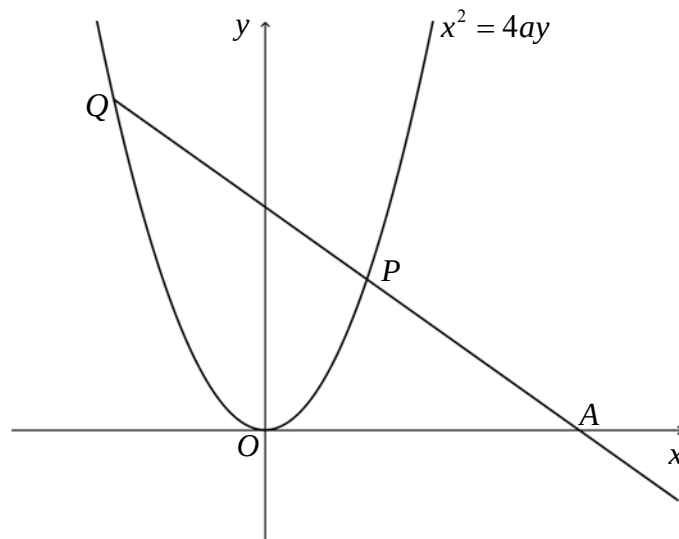
Question 12 (15 marks) Start a new writing booklet

- (a) The depth of water in a harbour varies during the day and is given by the equation

$d = 16 + 7 \sin\left(\frac{\pi}{12}t\right)$, where d is measured in metres and t is the number of hours after midnight.

- (i) Find the depth of the water at low and high tide. 1
- (ii) At what time(s) does high tide occur during the day? 1
- (iii) It is safe for a cruise ship to be in the harbour when the depth is more than 20 metres. 2
Find between what times during the day it is safe for the cruise ship to be in the harbour.
Give your answer correct to the nearest minute.

- (b) The chord QP joining the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ on $x^2 = 4ay$ always passes through the point A when produced.



NOT TO
SCALE

- (i) Show that the equation of the chord PQ is $(p + q)x - 2y = 2apq$. 2
- (ii) A has coordinates $(2a, 0)$. Show that $p + q = pq$. 1

Let $M(x, y)$ be the midpoint of QP .

- (iii) Show that the Cartesian Equation of the locus of M is a parabola with equation 2
 $(x - a)^2 = 2ay + a^2$.
- (iv) Hence, find the focus and directrix of the locus of M . 2

Question 12 continues on page 9

Question 12 (continued)

(c) Let $f(x) = \log_e(x-1) + \log_e 3$, for $x > 1$.

(i) Show that $f^{-1}(x) = \frac{e^x}{3} + 1$. **1**

(ii) Show that $y = f(x)$ and $y = f^{-1}(x)$ intersect when $3x - 3 = e^x$. **1**

(iii) Hence, show graphically, or otherwise, that $y = f(x)$ and $y = f^{-1}(x)$ do not intersect. **2**

End of Question 12

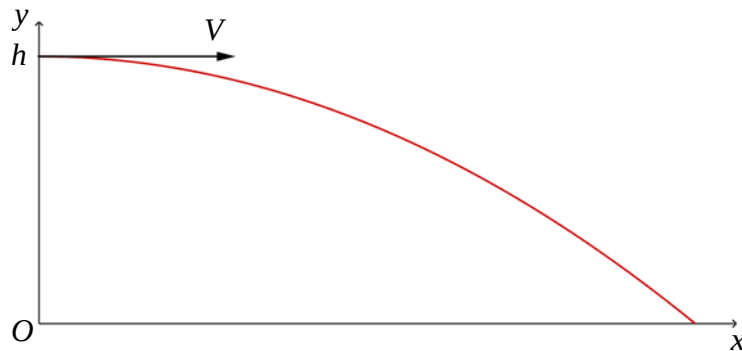
Question 13 (15 marks) Start a new writing booklet

- (a) 12 people need to be split up into teams for a quiz.
- (i) Explain why the number of ways of splitting them into two groups of the same size is $\frac{1}{2} {}^{12}C_6$. 1
 - (ii) How many ways are there of splitting them into two groups of any size (but there must be at least one person in each group)? 2
 - (iii) At the end of the quiz, the 12 people sit around a circular table to have dinner. In how many ways can they be seated? 1
- (b) Consider the polynomial $P(x) = x^3 + (p-2)x^2 + (5-2p)x - 10$.
- (i) Show that $(x-2)$ is a factor of $P(x)$. 1
 - (ii) By performing long division, show that $P(x) = (x-2)(x^2 + px + 5)$. 1
 - (iii) Find the exact values of p for which $P(x) = 0$ has only two distinct real roots. 2
 - (iv) For the larger of the two values of p found in (iii), sketch $y = P(x)$ showing the x - and y -intercept(s). 2

Question 13 continues on page 11

Question 13 (continued)

- (c) A particle is projected horizontally with velocity $V \text{ ms}^{-1}$, from a point h metres above the ground. Take $g = 10 \text{ ms}^{-2}$ as the acceleration due to gravity.



NOT TO
SCALE

Taking the origin as the point on the ground vertically below the point of projection, horizontal displacement is given by $x = Vt$. **Do not prove this result.**

- (i) Derive the expression for vertical displacement y . 1
- (ii) Show that the equation of the path is given by $y = h - \frac{5x^2}{V^2}$. 1
- (iii) Show that the range of the particle is $\frac{V\sqrt{h}}{\sqrt{5}}$ metres. 1
- (iv) If the projectile hits the ground at an angle of $\frac{\pi}{4}$ with the negative direction of the x -axis, find V in terms of h . 2

End of Question 13

Question 14 (15 marks) Start a new writing booklet

(a) The expansion of $(a+x)^n$ has the form $a^n + \dots + \alpha x^r + \beta x^{r+1} + \gamma x^{r+2} + \dots + x^n$, where n is a positive integer, a is a non-zero constant and α , β and γ are coefficients.

(i) Show that $\frac{\alpha}{\beta} = \frac{a(r+1)}{n-r}$. **2**

(ii) Show that if $a = 1$ and the expansion contains two consecutive terms with equal coefficients, then n will be odd. **1**

(iii) Using the result from (i), deduce an expression for $\frac{\beta}{\gamma}$. **1**

(iv) Prove that there are no values for a such that the expansion of $(a+x)^n$ has three consecutive terms with the same coefficient. **2**

Question 14 continues on page 13

Question 14 (continued)

(b) The function f is defined by $f(x) = e^{px}(x+1)$ where p is a non-zero real number.

(i) Show that $f'(x) = e^{px}[p(x+1)+1]$. 1

(ii) Let $f^{(n)}(x)$ denote the result of differentiating $f(x)$ with respect to x , n times. 2

Use mathematical induction to prove that for any positive integer n that

$$f^{(n)}(x) = p^{n-1}e^{px}[p(x+1)+n].$$

(iii) When $p = \sqrt{3}$, the graph of the function has exactly one point of inflexion. 1
Without proving a change in concavity, find the exact value of the x -coordinate of the point of inflexion.

Consider the function when $p = \frac{1}{2}$, that is $f(x) = e^{\frac{x}{2}}(x+1)$.

The graph of $y = e^{\frac{x}{2}}(x+1)$ has a minimum turning point at $\left(-3, \frac{-2}{e\sqrt{e}}\right)$, and a point of inflexion at $\left(-5, \frac{-4}{e^2\sqrt{e}}\right)$.

(iv) Sketch $y = e^{\frac{x}{2}}(x+1)$, showing the stationary point, the point of inflexion, the intercepts with the coordinate axes and any asymptotes. 2

It is known that $\frac{d}{dx}\left[\frac{e^{px}}{p}(x+1) - \frac{e^{px}}{p^2}\right] = e^{px}(x+1)$. **Do not prove this result.**

(v) Let R be the region enclosed by the curve $y = e^{\frac{x}{2}}(x+1)$, the x -axis and the lines $x = -2$ and $x = 2$. 3

Show that the area of R is $\frac{2(e^2 + 4\sqrt{e} - 3)}{e}$ square units.

End of Paper

HSC Mathematics Extension 1 Trial Examination 2017 HGHS Solutions

Multiple Choice

1.

$$y = \sin^{-1} \frac{x}{2}$$

$$-1 \leq \frac{x}{2} \leq 1$$

$$-2 \leq x \leq 2$$

(B)

2.

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} (2x^2 - 4)$$

$$= 4x$$

When $x = 2$, $\ddot{x} = 8$

(D)

3.

$$N = 80 + 20e^{kt}$$

Let $t = 0$

$$N = 80 + 20$$

$$100$$

Let $N = 1100$

$$1100 = 80 + 20e^{kt}$$

$$e^{kt} = \frac{1020}{20}$$

$$t = \frac{1}{k} \ln \frac{1020}{20}$$

$$= 19.65 \dots$$

$$= 19 \text{ hours } 40 \text{ minutes}$$

(C)

4.

$$-6 + -5 + 2 = -9$$

$$-6 \times -5 + 2 \times -5 + 2 \times -6 = 8$$

(A)

5.

$$3x + 2y - 7 = 0$$

$$m_1 = \frac{-3}{2}$$

$$2x - y + 1 = 0$$

$$m_2 = 2$$

$$\tan \theta = \left| \frac{2 + \frac{3}{2}}{1 + 2 \times \frac{-3}{2}} \right|$$
$$= \frac{7}{4}$$

$$\theta = 60^\circ \text{ (nearest degree)}$$

(C)

6.

$$y = 2 \cos^{-1} u, u = 3x$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{-2}{\sqrt{1-u^2}} \times 3$$

$$= \frac{-6}{\sqrt{1-(3x)^2}}$$

$$= \frac{-6}{\sqrt{1-9x^2}}$$

(C)

7.

$$\cos \frac{x}{2} = b$$

$$\frac{x}{2} = 2k\pi \pm \cos^{-1} b$$

$$x = 4k\pi \pm 2 \cos^{-1} b$$

(D)

8.

$$y_1 = -2, y_2 = 4$$

$$m = -1, n = 3$$

$$y = \frac{-1 \times 4 + 3 \times -2}{-2 + 3}$$

$$= \frac{-10}{1}$$

$$= -10$$

(A)

9.

$$y = \cos x - \sin x$$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right)$$

$$= \sqrt{2} \left(\cos \frac{\pi}{4} \cos x - \sin \frac{\pi}{4} \sin x \right)$$

$$= \sqrt{2} \left(\cos \left(x + \frac{\pi}{4} \right) \right)$$

By observation, (A) or (B) match amplitude

Check $x = 0$

$$y = \sqrt{2} \left(\cos \frac{\pi}{4} \right)$$

$$= 1$$

(A)

10.

$$\int \frac{1}{x^2 + 2x + 3} dx = \int \frac{1}{2 + (x+1)^2} dx$$

$$= \frac{1}{\sqrt{2}} \tan \left(\frac{x+1}{\sqrt{2}} \right) + C$$

(C)

Question 11

(a)

$$\frac{2}{x-1} \geq x \quad x \neq 1$$

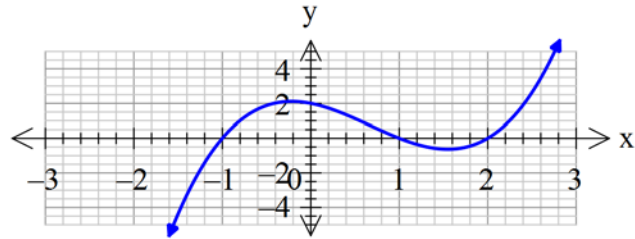
$$2(x-1) \geq x(x-1)^2$$

$$0 \geq x(x-1)^2 - 2(x-1)$$

$$(x-1)[x(x-1)-2] \leq 0$$

$$(x-1)(x^2 - x - 2) \leq 0$$

$$(x-1)(x-2)(x+1) \leq 0$$



From graph $x \leq -1$ or $1 < x \leq 2$

(b)

$$\cos 2x = 2 \cos^2 x - 1$$

$$\frac{1}{2}(\cos 2x + 1) = \cos^2 x$$

$$\cos^2 3x = \frac{1}{2}(\cos 6x + 1)$$

$$\int_0^\pi \cos^2 3x dx = \frac{1}{2} \int_0^\pi (\cos 6x + 1) dx$$

$$= \frac{1}{2} \left[\frac{1}{6} \sin 6x + x \right]_0^\pi$$

$$= \frac{1}{2} \left[\left(\frac{1}{6} \sin 6\pi + \pi \right) - \left(\frac{1}{6} \sin 0 + 0 \right) \right]$$

$$= \frac{1}{2} \times \pi$$

$$= \frac{\pi}{2}$$

(c)

(i)

$$\left(x + \frac{2}{x} \right)^5 = x^5 + 5x^4 \times \frac{2}{x} + 10x^3 \times \frac{4}{x^2} + 10x^2 \times \frac{8}{x^3} + 5x \times \frac{16}{x^4} + \frac{32}{x^5}$$

$$= x^5 + 10x^3 + 40x + \frac{80}{x} + \frac{80}{x^3} + \frac{32}{x^5}$$

(ii)

$$\left(x - \frac{2}{x} \right)^5 = x^5 - 10x^3 + 40x - \frac{80}{x} + \frac{80}{x^3} - \frac{32}{x^5}$$

$\therefore \left(x + \frac{2}{x}\right)^5 + \left(x - \frac{2}{x}\right)^5 = 2\left(x^5 + 40x + \frac{80}{x^3}\right)$ $= 2x^5 + 80x + \frac{160}{x^3}$	
(d) Let $u = e^x$ $\frac{du}{dx} = e^x$ $du = e^x dx$ When $x = 0$, $u = 1$ When $x = 1$, $u = e$ $\int_0^1 \frac{e^x}{\sqrt{e^x + 1}} dx = \int_1^e (u + 1)^{\frac{-1}{2}} du$ $= 2 \left[\sqrt{u + 1} \right]_1^e$ $= 2 \left(\sqrt{e + 1} - \sqrt{1 + 1} \right)$ $= 2\sqrt{e + 1} - 2\sqrt{2}$	
(e) Let $\angle ABD = \alpha$ $\angle BCE = \angle ABD$ (corresponding $\angle$'s, $BD \parallel CE$) $= \alpha$ $\angle DEB = \angle ABD$ ($\angle$ in the alternate segment) $= \alpha$ $\therefore \angle BCE = \angle BED$	

Question 12	
(a) $d = 16 + 7 \sin\left(\frac{\pi}{12}t\right)$ (i) Max depth is $16+7 = 23$ m, minimum depth 9 m	
(ii) Let $d = 23$ $23 = 16 + 7 \sin\left(\frac{\pi}{12}t\right)$ $7 = 7 \sin\left(\frac{\pi}{12}t\right)$ $\sin\left(\frac{\pi}{12}t\right) = 1$ $\frac{\pi}{12}t = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \dots$ $t = 6, 30, \dots$ 6:00am is the only time of the day.	
(iii) Let $d = 20$ $20 = 16 + 7 \sin\left(\frac{\pi}{12}t\right)$ $\frac{4}{7} = \sin\left(\frac{\pi}{12}t\right)$ $\frac{\pi}{12}t = \sin^{-1}\frac{4}{7}, \pi - \sin^{-1}\frac{4}{7}$ $t = \frac{12}{\pi} \sin^{-1}\frac{4}{7}, \frac{12}{\pi} \left(\pi - \sin^{-1}\frac{4}{7}\right)$ = 2 hours 19 minutes, 9 hours 41 minutes Therefore, between 2:19 am and 9:41 am.	
(b) (i) $P(2ap, ap^2) \quad Q(2aq, aq^2)$ $m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$ $= \frac{a(p-q)(p+q)}{2a(p-q)}$ $= \frac{p+q}{2}$ Equation of PQ:	

$y - ap^2 = \frac{p+q}{2}(x - 2ap)$ $2y - 2ap^2 = (p+q)(x - 2ap)$ $2y - 2ap^2 = x(p+q) - 2ap(p+q)$ $2y - 2ap^2 = x(p+q) - 2ap^2 - 2apq$ $x(p+q) - 2y = 2apq$	
(ii) Sub $x = 2a$, $y = 0$ into equation of chord $2a(p+q) = 2apq$ $p+q = pq$	
(iii) Consider the midpoint $M(x, y)$ $x = a(p+q)$ $\therefore \frac{x}{a} = p+q$ $y = \frac{ap^2 + aq^2}{2}$ $y = \frac{a(p^2 + q^2)}{2}$ $= \frac{a}{2}[(p+q)^2 - 2pq]$ $= \frac{a}{2}\left[\frac{x^2}{a^2} - 2(p+q)\right]$ $= \frac{a}{2}\left[\frac{x^2}{a^2} - \frac{2x}{a}\right]$ $\frac{2y}{a} = \frac{x^2 - 2xa}{a^2}$ $2ay = x^2 - 2xa$ $2ay + a^2 = x^2 - 2xa + a^2$ $(x-a)^2 = 2a\left(y + \frac{a}{2}\right)$	
(iv) Vertex $\left(a, -\frac{a}{2}\right)$, Focal length $\frac{a}{2}$, Focus $(a, 0)$, Directrix $y = -a$	

(c)

$$f(x) = \ln(x-1) + \ln 3$$

(i)

Finding inverse:

$$x = \ln(y-1) + \ln 3$$

$$x = \ln[3(y-1)]$$

$$e^x = 3(y-1)$$

$$\frac{e^x}{3} = y-1$$

$$y = 1 + \frac{e^x}{3}$$

$$f^{-1}(x) = \frac{e^x}{3} + 1$$

(ii)

$y = f^{-1}(x)$ and $y = f(x)$ intersect on $y = x$

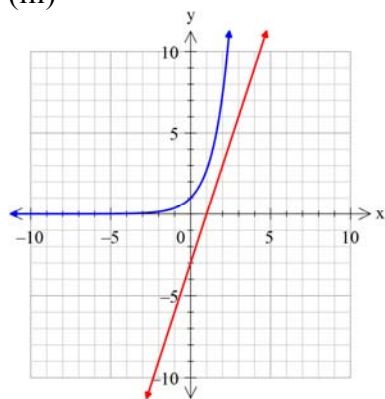
Let $f^{-1}(x) = x$

$$\frac{e^x}{3} + 1 = x$$

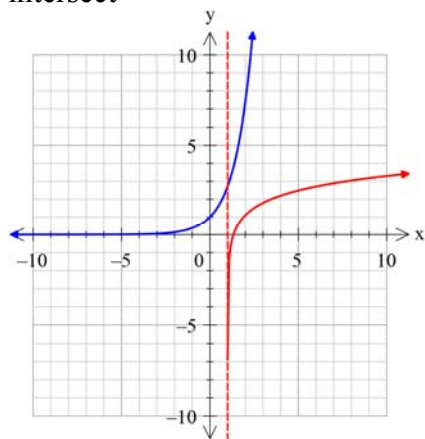
$$e^x + 3 = 3x$$

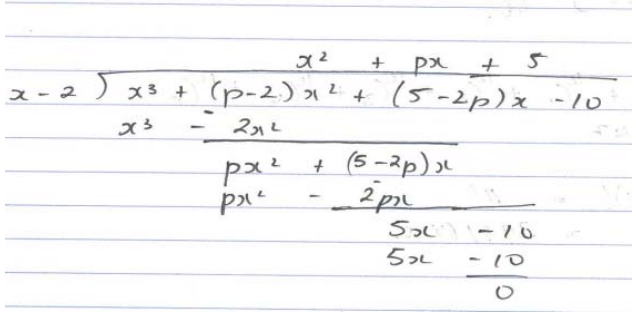
$$3x - 3 = e^x$$

(iii)

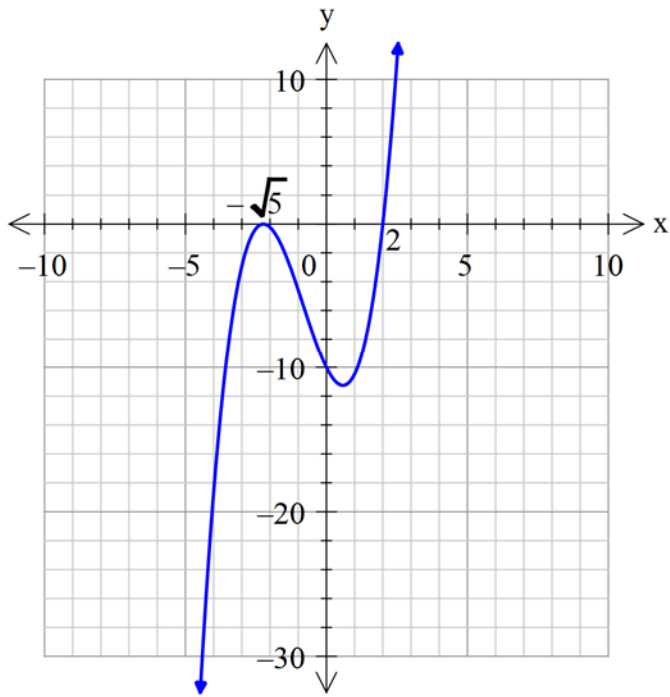


Therefore $y = f(x)$ and $y = f^{-1}(x)$ do not intersect



Question 13	
(a) (i) We want to select 6 from 12, ${}^{12}C_6$. When we do this, it leaves out a group of 6 (a team). Therefore for each combination we are counting twice the number of teams. $\therefore \frac{1}{2} \times {}^{12}C_6$	
(ii) Number of groups: $\frac{1}{2} \times \sum_{r=1}^{11} {}^{12}C_r = 2047$ Or $\frac{1}{2} {}^{12}C_6 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 + {}^{12}C_4 + {}^{12}C_5$	
(iii) $(12-1)! = 11!$ $= 39916800$	
(b) $P(x) = x^3 + (p-2)x^2 + (5-2p)x - 10$ (i) $P(2) = 2^3 + (p-2) \times 2^2 + (5-2p) \times 2 - 10$ $= 8 + 4(p-2) + 2(5-2p) - 10$ $= 8 + 4p - 8 + 10 - 4p - 10$ $= 0$ $\therefore (x-2)$ is a factor of $P(x)$	
(ii)  $\therefore (x-2)(x^2 + px + 5) = P(x)$	
(iii) $x=2$ is a root. We need one root from $x^2 + px + 5 = 0$ $\Delta = p^2 - 4 \times 5 \times 1$ $= p^2 - 20$ Let $\Delta = 0$ $p^2 - 20 = 0$ $p = \pm \sqrt{20}$ $= \pm 2\sqrt{5}$	

Let $p = 2\sqrt{5}$



(c)

(i)

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c_1$$

When $t = 0$,

$$\dot{y} = V \sin 0^\circ$$

$$= 0$$

$$\therefore c_1 = 0$$

$$\dot{y} = -gt$$

$$y = -\frac{gt^2}{2} + c_2$$

When $t = 0$, $y = h$

$$h = 0 + c_2$$

$$c_2 = h$$

$$\therefore y = -\frac{gt^2}{2} + h$$

$$= -5t^2 + h$$

(ii)

$$x = Vt$$

$$t = \frac{x}{V}$$

Substitute into y :

$$y = -10\left(\frac{x}{V}\right)^2 + h$$

$$= h - 5 \times \frac{x^2}{V^2}$$

$$= h - \frac{5x^2}{V^2}$$

(iii)

Let $y = 0$

$$0 = h - \frac{5x^2}{V^2}$$

$$\frac{5x^2}{V^2} = h$$

$$5x^2 = hV^2$$

$$x^2 = \frac{h}{5}V^2$$

$$x = V\sqrt{\frac{h}{5}}$$

$$= V\frac{\sqrt{h}}{\sqrt{5}}$$

(iv)

$$\frac{dy}{dx} = \frac{-10x}{V^2}$$

When $x = \frac{V\sqrt{h}}{\sqrt{5}}$,

$$\frac{dy}{dx} = \frac{-10}{V^2} \times \frac{V\sqrt{h}}{\sqrt{5}}$$

$$= \frac{-10\sqrt{h}}{\sqrt{5}V}$$

But

$$\frac{dy}{dx} = \tan 135^\circ$$

$$= -1$$

$$\therefore \frac{-10\sqrt{h}}{\sqrt{5}V} = -1$$

$$-10\sqrt{h} = -\sqrt{5}V$$

$$\sqrt{5}V = \sqrt{5}\sqrt{2}\sqrt{10}\sqrt{h}$$

$$V = \sqrt{20h}$$

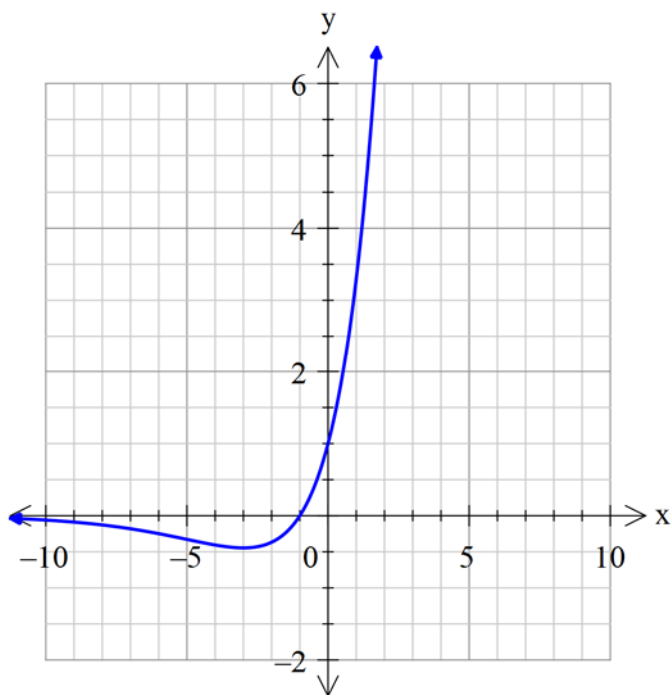
Question 14	
(a) $(a+x)^n = \sum_{k=0}^n {}^nC_k a^{n-k} x^k$ (i) $\frac{\alpha}{\beta} = \frac{{}^nC_r a^{n-r}}{{}^nC_{r+1} a^{n-(r+1)}}$ $= \frac{n!}{(n-r)!r!} \div \frac{n!}{(n-r-1)!(r+1)!} \times \frac{a^{n-r}}{a^{n-r-1}}$ $= a \times \frac{n!}{(n-r)!r!} \times \frac{(n-r-1)!(r+1)!}{n!}$ $= \frac{a(r+1)}{n-r}$	
(ii) Let $a=1$ and $\frac{\alpha}{\beta} = 1$ $1 = \frac{r+1}{n-r}$ $n-r = r+1$ $n = 2r+1$ Since r is a non-negative integer, $2r+1$ will be an odd number. Therefore there will be two consecutive coefficients with n is odd.	
(iii) $\frac{\beta}{\gamma} = \frac{a((r+1)+1)}{n-(r+1)}$ $= \frac{a(r+2)}{n-r-1}$	
(iv) For three consecutive equal coefficients: $\frac{\alpha}{\beta} = \frac{\beta}{\gamma} = 1$ $\frac{a(r+2)}{n-r-1} = \frac{a(r+1)}{n-r}$ $\frac{r+2}{n-r-1} = \frac{r+1}{n-r}$ $(r+2)(n-r) = (n-r-1)(r+1)$ $nr - r^2 + 2n - 2r = nr + n - r^2 - r - r - 1$ $2n - 2r = n - 2r - 1$ $2n = n - 1$ $n = -1$ But n is a positive integer. Therefore there will be no values of a such that three consecutive terms have the same coefficient.	
(b) $f(x) = e^{px}(x+1)$	

(i) $f'(x) = pe^{px}(x+1) + 1 \times e^{px}$ $= e^{px}(p(x+1) + 1)$	
(ii) Let $n = 1$ Proven above Assume true for $n = k$ $f^{(k)}(x) = p^{k-1}e^{px}(p(x+1) + k)$ Let $n = k + 1$ RTP: $f^{(k+1)}(x) = p^k e^{px}(p(x+1) + (k+1))$ $f^{(k+1)}(x) = \frac{d}{dx}(f^{(k)}(x))$ $= p^{k-1} \times pe^{px}(p(x+1) + k) + p^{k-1}e^{px} \times p$ $= p^k e^{px}(p(x+1) + k) + p^k e^{px}$ $= p^k e^{px}(p(x+1) + k + 1)$ As required. $\therefore$ True for $n = k + 1$ if true for $n = k$ By the principle of mathematical induction, the statement is true for integer $n \geq 1$.	
(iii) Let $p = \sqrt{3}$ $f(x) = e^{\sqrt{3}x}(x+1)$ $f''(x) = \sqrt{3}e^{\sqrt{3}x}(\sqrt{3}(x+1) + 2)$ Let $f''(x) = 0$ $\sqrt{3}e^{\sqrt{3}x}(\sqrt{3}(x+1) + 2) = 0$ $\sqrt{3}(x+1) + 2 = 0$ $x+1 = \frac{-2}{\sqrt{3}}$ $x = -1 - \frac{2}{\sqrt{3}}$ $= -\left(\frac{3+2\sqrt{3}}{3}\right)$	
(iv) $f(x) = e^{\frac{x}{2}}(x+1)$ Let $f(x) = 0$	

$$e^{\frac{x}{2}}(x+1)=0$$

$$x+1=0 \quad \left(e^{\frac{x}{2}} > 0 \right)$$

$$x=-1$$



(v)

$$A = - \int_{-2}^{-1} e^{\frac{x}{2}}(x+1) dx + \int_{-1}^2 e^{\frac{x}{2}}(x+1) dx$$

$$= \left[2e^{\frac{x}{2}}(x+1) - 4e^{\frac{x}{2}} \right]_{-1}^{-2} + \left[2e^{\frac{x}{2}}(x+1) - 4e^{\frac{x}{2}} \right]_{-1}^2$$

$$= 2 \left(\left[e^{\frac{x}{2}}(x+1-2) \right]_{-1}^{-2} + \left[e^{\frac{x}{2}}(x+1-2) \right]_{-1}^2 \right)$$

$$= 2 \left(\left[e^{\frac{x}{2}}(x-1) \right]_{-1}^{-2} + \left[e^{\frac{x}{2}}(x-1) \right]_{-1}^2 \right)$$

$$= 2 \left[\left(e^{-1} \times -3 \right) - \left(e^{\frac{-1}{2}} \times -2 \right) + \left(e \times 1 \right) - \left(e^{\frac{-1}{2}} \times -2 \right) \right]$$

$$= 2 \left(\frac{-3}{e} + \frac{2}{\sqrt{e}} + e + \frac{2}{\sqrt{e}} \right)$$

$$= 2 \left(e + \frac{4}{\sqrt{e}} - \frac{3}{e} \right)$$

$$= 2 \left(\frac{e^2 + 4\sqrt{e} - 3}{e} \right)$$